BIOT 6900 · Module 3: Target Triage with Protein Language Models

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1. Re-ranking

Did the prior agree with my Module 2 ranking? Mostly not, and that is expected. The two scores measure different things (disease association versus resemblance to drugged proteins), and the Spearman correlation between my Module 2 score and the best-modality prior is only **0.21**. The small positive value comes from the antibody side (0.38 with p_ab, −0.16 with p_sm): my strongest Module 2 hits are immune and cell-surface proteins. The targets fall into three groups:

- **Looks druggable:** PECAM1 (p_ab 0.88), APLNR (p_sm 0.68), ITGAX (p_ab 0.59), MOG (p_ab 0.50)
- **In between (0.1 to 0.5):** NRXN3 (p_sm 0.44), SHANK2 (p_sm 0.33), HLA-DRB1 (p_ab 0.27), APOC1 (p_ab 0.25), HLA-DRA (p_ab 0.18), GMPR (p_sm 0.17), APOB (p_sm 0.15), MTSS2 (p_sm 0.12)
- **No precedent for either modality (below 0.1):** RPH3A, APOE, PLEC

Final	Gene	Module 2	Change	p_sm	p_ab	Nearest drugged proteins (cosine)
1	PECAM1	4	+3	0.02	0.88	CD2 (0.96), IL12B (0.96), LAMA4 (0.95)
2	HLA-DRB1	2	0	0.01	0.27	SIGLEC8 (0.87), IFNA10 (0.86), PTGDS (0.85)
3	HLA-DRA	1	−2	0.01	0.18	SEMA4D (0.92), GRIK3 (0.92), CA12 (0.91)
4	ITGAX	6	+2	0.21	0.59	ITGA2B (0.98), CSF1R (0.96), SEMA4D (0.95)
5	APOC1	3	−2	0.00	0.25	IFNA5 (0.86), SCTR (0.85), SLC34A2 (0.85)
6	APLNR	10	**+4**	0.68	0.03	AGTR2 (0.96), GHSR (0.95), S1PR2 (0.94)
7	NRXN3	9	+2	0.44	0.12	SDC4 (0.94), KCNB1 (0.93), RELA (0.92)
8	MTSS2	5	−3	0.12	0.01	PDE4D (0.94), KCNQ4 (0.93), PDE4C (0.93)
9	GMPR	7	−2	0.17	0.00	EEF2 (0.95), PSMA5 (0.93), RPL11 (0.93)
10	SHANK2	12	+2	0.33	0.00	E2F7 (0.97), MAP3K13 (0.96), KCNQ3 (0.95)
11	MOG	14	+3	0.00	0.50	PDCD1LG2 (0.93), TIGIT (0.92), IFNGR2 (0.91)
12	RPH3A	8	**−4**	0.09	0.00	CACNB3 (0.93), PDE4D (0.93), CACNB2 (0.92)
13	APOE	11	−2	0.00	0.01	MADCAM1 (0.86), TNFSF14 (0.85), IL11RA (0.84)
14	APOB	13	−1	0.15	0.10	AFP (0.97), LAMA4 (0.95), IFNAR2 (0.95)
15	PLEC	15	0	0.01	0.00	LAMB2 (0.89), EPHA10 (0.87), EPHA1 (0.86)

APOE and APOB tie on the final score (0.253); the tie is broken by Module 2 order.

Which target moved most, and why? Two targets moved four places:

- **RPH3A fell furthest (8th to 12th):** Rabphilin-3A is a synaptic-vesicle protein inside the cell that works by binding other proteins and membranes rather than through a drug-like pocket. Antibodies cannot reach it (p_ab 0.00), and its small-molecule prior is low (0.09). Its nearest drugged proteins, the calcium-channel subunits CACNB3 and CACNB2, are most likely labelled drugged because the channel they belong to is a drug target, not because they have a pocket themselves
- **APLNR rose furthest (10th to 6th):** it is the apelin receptor, a GPCR, and its nearest drugged proteins (AGTR2, GHSR, S1PR2) are all GPCRs of the same class, the classic small-molecule target family

PECAM1 and MOG also rose (+3 each) on strong antibody priors.

Justifying w = 0.6: I weighted Module 2 evidence slightly above tractability:

1. **Disease relevance comes first:** a druggable protein that does not cause AD is a wasted programme, and tractability can be worked around (a different modality or a partner in the same pathway) while causality cannot
2. **The prior is only moderately reliable:** its family-split AUROC is 0.745 (Section 3), so it should adjust the ranking, not override it
3. **Sensitivity:** at w = 0.5 the prior starts to override strong genetics (HLA-DRA, the Module 2 #1, falls to 4th behind ITGAX). At w = 0.7 the top six are exactly my Module 2 top six, so tractability only reshuffles within them and APLNR, my best small-molecule target, drops to 7th

w = 0.6 keeps the disease signal in charge while still moving clearly tractable targets up.

2. Modality

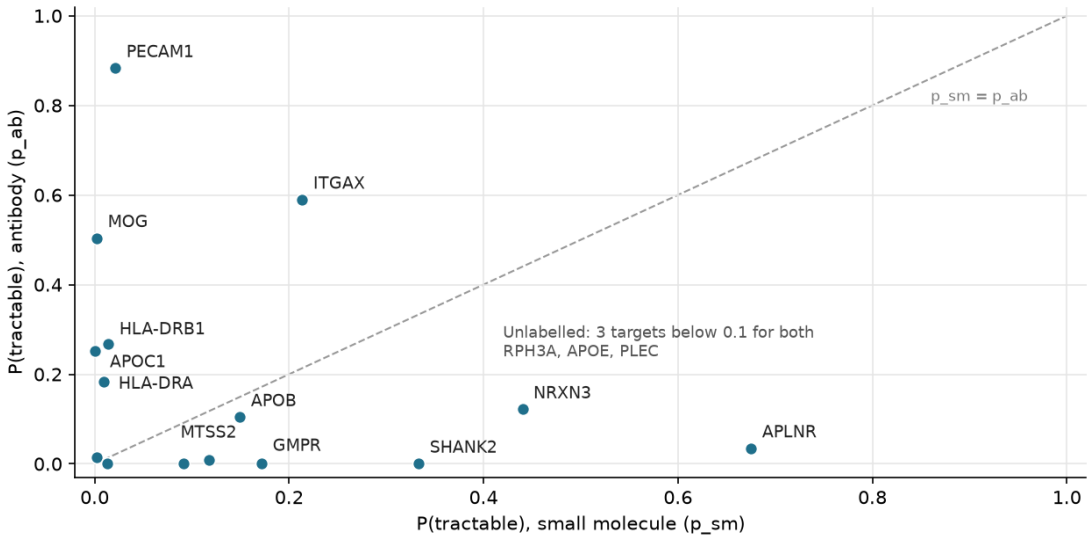


Figure 1: Modality map of my 15 targets. Cell-surface and secreted targets (PECAM1, ITGAX, MOG) sit high on the antibody axis; only the GPCR APLNR sits far out on the small-molecule axis; RPH3A, APOE and PLEC are near zero for both.

Target: PECAM1 (p_ab 0.88, p_sm 0.02), the sharpest antibody-over-small-molecule split.

- **Location:** a membrane protein on blood-vessel lining cells, platelets and white blood cells, with a large outer region exposed to the blood
- **Family:** made of six antibody-like (immunoglobulin) domains. Its nearest drugged proteins are a cell-surface receptor (CD2), a secreted cytokine subunit (IL12B) and a laminin (LAMA4), all outside the cell, and CD2 and IL12B are antibody-drugged
- **Likely pocket:** none deep, since these domains bind partners (PECAM1 binds itself between cells) through flat, broad surfaces that small molecules cannot grip
- **AD angle:** PECAM1 sits on the blood vessels of the blood–brain barrier, so an antibody can reach it from the blood without having to cross into the brain

The mirror image, APLNR (p_sm 0.68, p_ab 0.03): a GPCR that crosses the membrane seven times. Its natural ligand binds in a pocket inside this bundle, which is where most GPCR small-molecule drugs bind, and its nearest drugged proteins are three other GPCRs of the same class. The part an antibody could reach outside the cell is small. The panel has 53 GPCRs of this class, 33 of them with clinical precedence, so this prior rests on a well-populated family.

The hard case, APOE (p_sm 0.00, p_ab 0.01): a secreted lipid carrier with no enzyme pocket or receptor shape, so it looks like neither drug class. That is a modality problem, not a reason to drop it. Options include small molecules that correct the shape of ApoE4, antibodies against ApoE stuck in amyloid plaques, or targeting the proteins it works with (LDLR, ABCA1).

3. Validation

Split	AUROC	Average precision
Random	0.839	0.717
Family (held out by family group)	**0.745**	**0.548**
Chance	0.500	0.333

Why the family split is the honest number: in a random split, close relatives sit on both sides of a fold. If AKT1 is in training, AKT3 is easy to "predict". That rewards memorising families, not learning what makes a protein tractable. In this panel the family split is strict: all 61 kinases form one group, and all 53 GPCRs of the main class another, so each family is held out as a whole. That matches the real situation for a new AD target whose family may never have been drugged (APOC1, MTSS2, RPH3A).

What the random split would have hidden: a drop of 0.094 AUROC and 0.169 AP. The AP gain over chance shrinks from 0.384 to 0.215, so the random split hides about 44% of it. The prior still clearly beats chance, but a team reading 0.84 would trust it more than it deserves, especially for proteins unlike anything in the panel.

4. Variants

I number residues as UniProt does, which counts APOE's 18-residue signal peptide, so the positions are 18 higher than the usual ApoE names (R176C is the familiar R158C, ε2, and C130R is C112R, ε4). Scores come from ESM-2 650M and compare how likely the model finds the mutant versus the original residue (log P(mutant) − log P(wild type)). The percentile ranks each variant among every possible substitution in that protein, so a low percentile means more surprising than most.

Gene	Variant (UniProt)	Meaning	Score	Percentile in protein
APOE	R176C (rs7412)	ε2, protective for AD	−8.56	11.8
APOE	C130R (rs429358)	ε4, strongest common AD risk allele	**+7.21**	**100.0**
TREM2	R47H (rs75932628)	rare AD risk, impairs ligand binding	−5.16	47.7
TP53	R175H	cancer hotspot (structural)	−5.97	19.3

For comparison, the median substitution scores −3.9 in APOE, −4.8 in TREM2 and −2.2 in TP53.

Interpreting the scores:

- **APOE ε4 (C130R), the most "expected" change in all of APOE:** the model prefers arginine over the human reference cysteine here, because ε4 is the ancestral version and other mammals carry arginine at this position, yet it is the largest common genetic risk factor for AD
- **APOE ε2 (R176C), scored as damaging:** R176 sits in the region that binds the LDL receptor, and ε2 does weaken that binding (two copies cause a cholesterol disorder), so the model flags a real functional change that happens to protect against AD
- **TREM2 R47H, middling:** no more surprising than a typical TREM2 substitution despite a two- to fourfold rise in AD risk, because it subtly weakens ligand binding on the protein surface rather than breaking the fold
- **p53 R175H, in the damaging tail:** expected for a hotspot that breaks the fold, but not the most extreme hotspot (R248Q scores −9.55, 5.2nd percentile), since cancer selects for mutations that change what p53 does, not for how unusual they look in evolution

The two APOE alleles show the key point: the model ranks them in the **opposite order** to their AD risk.

Why a strongly negative score is not "pathogenic": the score measures how unusual a change is given patterns learned from millions of natural sequences. That is evolutionary surprise, not disease. A change can be surprising and harmless, or ordinary and disease-causing (like ε4).

What these models cannot see:

- **Variant types:** changes outside the protein-coding sequence, splicing changes, insertions and deletions, and copy-number changes
- **Context:** binding partners (lipid, receptor, DNA), cell type, expression level, dose (one or two copies), age, and interactions with other variants

- **Clinical evidence:** population frequency, family and patient data used by ClinVar and clinical classification rules, which Week 5 compares against CADD, REVEL and SpliceAI

5. Limitations and next step

What sequence alone cannot tell me:

- **Pockets and 3D shape:** the prior says "looks like drugged proteins", not "has a pocket a drug can bind", and hidden or mutant-only pockets (p53 Y220C) are invisible
- **History in the labels:** labels record where industry has worked, so undrugged classes (apolipoproteins, scaffolds such as SHANK2 and PLEC) score low partly because nobody has tried. Labels can also pass to every subunit of a drugged complex, which is likely why two of GMPR's nearest "drugged" proteins are a proteasome subunit and a ribosomal protein (PSMA5, RPL11)
- **An unbalanced antibody prior:** only 66 training proteins have antibody precedence, against 301 for small molecules, so p_ab rests on far fewer examples than p_sm
- **Brain delivery:** nothing here models the blood–brain barrier, which decides whether a drug reaches a target on brain immune cells such as ITGAX or HLA-DR
- **Direction and safety:** whether to inhibit or activate, and side effects (HLA-DR is central to immune function)
- **Similarity is not shared ancestry:** a typical panel pair already has cosine 0.72 (5th to 95th percentile 0.30 to 0.90), so a top-3 match at 0.85 to 0.98 is a ranking, and some matches (SHANK2 → E2F7, APOB → AFP, both 0.97) likely reflect protein size and composition rather than shared biology

Two targets for Week 4 (structure-based pocket assessment):

1. **ITGAX (final rank 4, p_ab 0.59):** CD11c marks the brain immune cells that gather around AD damage, so it is close to AD biology. ITGAX has a ligand-binding domain that is drugged in its relative ITGAL by lifitegrast, and that domain also has a second pocket that experimental inhibitors bind. Week 4 should test whether ITGAX's version shows similar pockets, which would open a small-molecule route alongside antibodies
2. **APLNR (final rank 6, p_sm 0.68):** my strongest small-molecule prior, a GPCR with a defined ligand pocket, and experimental apelin-receptor structures exist, so Week 4 can check the predicted structure against them before assessing the pocket

I did not pick PECAM1, despite its first place: it is an antibody target with flat binding surfaces, so pocket assessment would add little. I would hold APOE as an exploratory third structure, because ApoE4 has a known site where shape-correcting drugs bind, which a sequence prior cannot see.

Unmapped genes

Unmapped targets in my top 15: none. All 15 have a reviewed UniProt protein, so all were scored. In my full Module 2 list of 1,115 genes the only gene without a protein is **UBBP4**, a ubiquitin pseudogene. A protein language model cannot score it. It would need different evidence:

- **Expression evidence:** RNA-seq showing whether the pseudogene is made into RNA in AD brain
- **Regulatory evidence:** whether its RNA has a role of its own (for example soaking up microRNAs that would otherwise act on UBB), tested by knocking it down
- **Genetic evidence:** whether variants at its location affect its expression and line up with AD risk signals